

Machine Learning

Support Vector Machine

FAST

DISCOVERING
THE FUTURE

Topics of previous lectures

- ✓ Ingredients of Machine Learning
- ✓ Classification Basics
- ✓ Basic Linear Classifier
- ✓ K-Nearest Neighbours Classifier
- ✓ Naive Bayes Classifier
- ✓ Decision Trees

Topics of today's lecture

- Convex Optimization
- Support Vector Machine (SVM)
- Hard-margin SVM
- Soft-margin SVM
- Kernel methods

Background for Constrained Optimization

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$$\min_{x,y} f(x,y)$$

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where $\lambda \in \mathbb{R}$ is the Lagrange multiplier

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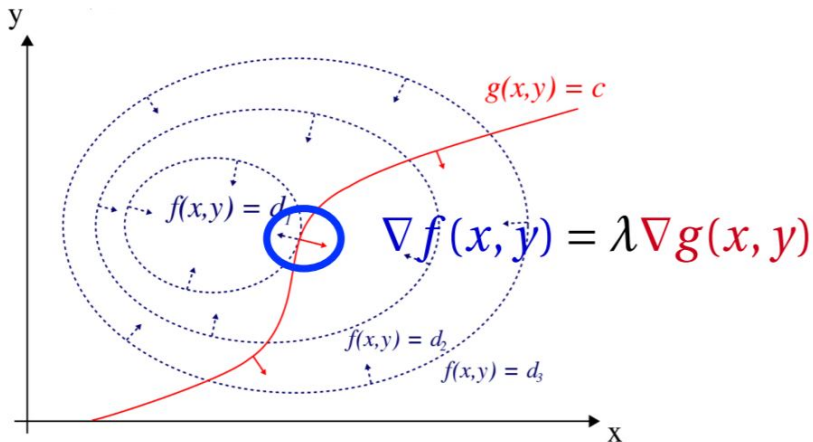
where $\lambda \in \mathbb{R}$ is the Lagrange multiplier

- Solve the unconstrained problem: $\nabla \Lambda(x,y,\lambda) = 0$, which is equivalent to

$$\nabla_{x,y} \Lambda(x,y,\lambda) = \nabla f(x,y) - \lambda \nabla g(x,y) = 0$$

$$\nabla_{\lambda} \Lambda(x,y,\lambda) = g(x,y) - c = 0$$

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The following optimization problem

$$\min_{\mathbf{x}} f(\mathbf{x})$$

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The associated Lagrangian **dual problem** is given by

$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^m} \min_{\mathbf{x} \in \mathbb{R}^d} \Lambda(\mathbf{x}, \boldsymbol{\lambda})$$

$$\text{subject to } \boldsymbol{\lambda} \geq 0$$

where $\boldsymbol{\lambda}$ are the *dual variables*.

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Definition (Convex set)

A set C is a convex set if $\forall x, y \in C$ and $\forall \theta \in [0, 1]$, we have

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Definition (Convex function)

Let $f : X \rightarrow \mathbb{R}$ be a function such that X is a convex set, then f is a convex (concave) function if $\forall \mathbf{x}_1, \mathbf{x}_2 \in X$ and $\forall \theta \in [0, 1]$, we have

$$f(\theta \mathbf{x}_1 + (1 - \theta)\mathbf{x}_2) \leq \theta f(\mathbf{x}_1) + (1 - \theta)f(\mathbf{x}_2)$$

Theorem 1

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable, then $f(\mathbf{x})$ is convex iff $\forall \mathbf{x}_1, \mathbf{x}_2$

$$f(\mathbf{x}_2) \geq f(\mathbf{x}_1) + \nabla_{\mathbf{x}} f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1).$$

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Now let's look at two well-known classes of convex optimization problems.

Consider the case when the objective function is linear

$$\min_{\mathbf{x} \in \mathbb{R}^d} \mathbf{c}^T \mathbf{x}$$

subject to $\mathbf{Ax} \leq \mathbf{b}$,

where $\mathbf{A} \in \mathbb{R}^{m \times d}$, $\mathbf{c} \in \mathbb{R}^d$ and $\mathbf{b} \in \mathbb{R}^m$

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The Lagrangian will be

$$\Lambda(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{c}^T \mathbf{x} + \boldsymbol{\lambda}^T (\mathbf{A}\mathbf{x} - \mathbf{b}) = (\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda})^T \mathbf{x} - \boldsymbol{\lambda}^T \mathbf{b},$$

where $\boldsymbol{\lambda} \in \mathbb{R}^m$ is the vector of non-negative Lagrange multipliers.

Linear Programming

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The dual Lagrangian problem will be

$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^m} -\boldsymbol{\lambda}^T \mathbf{b}$$

subject to $\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda} = 0, \quad \boldsymbol{\lambda} \geq 0$

Quadratic Programming

Consider the case when the objective function is quadratic

$$\min_{\mathbf{x} \in \mathbb{R}^d} \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x}$$

subject to $\mathbf{A} \mathbf{x} \leq \mathbf{b}$,

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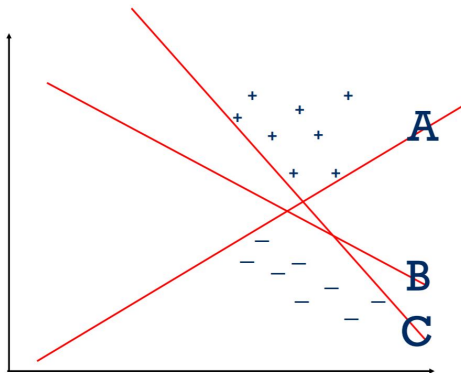
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The dual Lagrangian problem will be

$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^m} -\frac{1}{2} (\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda})^T \mathbf{Q}^{-1} (\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda}) - \boldsymbol{\lambda}^T \mathbf{b}$$

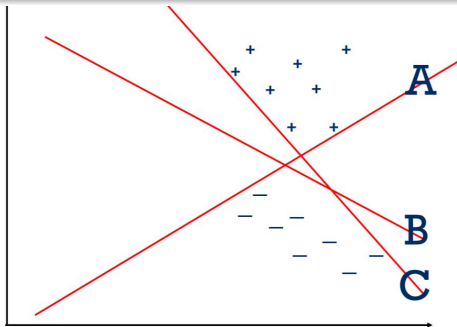
subject to $\boldsymbol{\lambda} \geq 0$

Motivation for SVM



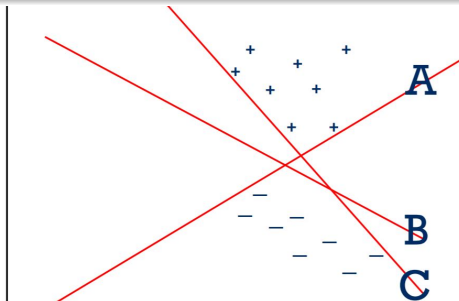
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There are infinitely many separating classifiers (decision boundaries) between linearly separable classes. Which one is the best in your opinion?



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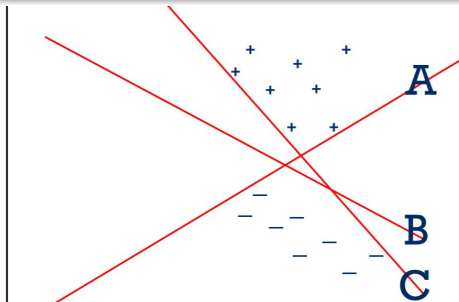
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Support Vector Machine learns the separating line B.

Support Vector Machine (SVM)

- Again a linear classification method

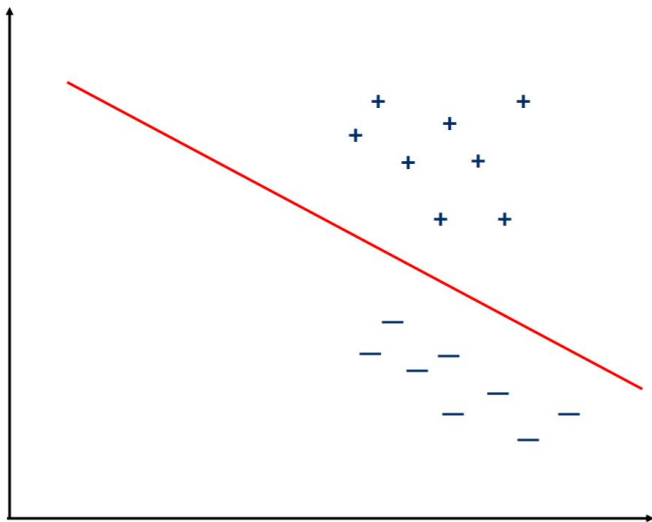
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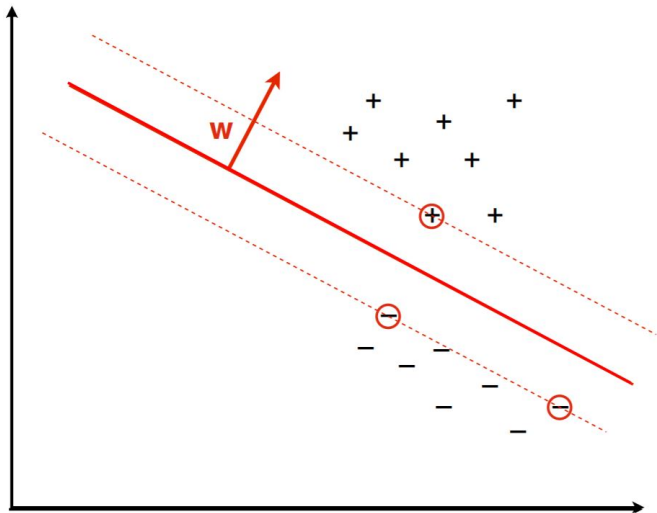
Support Vector Machine (SVM)

- Again a linear classification method
- If the classes are linearly separable then SVM finds a separating model
- SVM chooses the linear separating model which has the highest **margin** - distance between the decision boundary and the closest instance

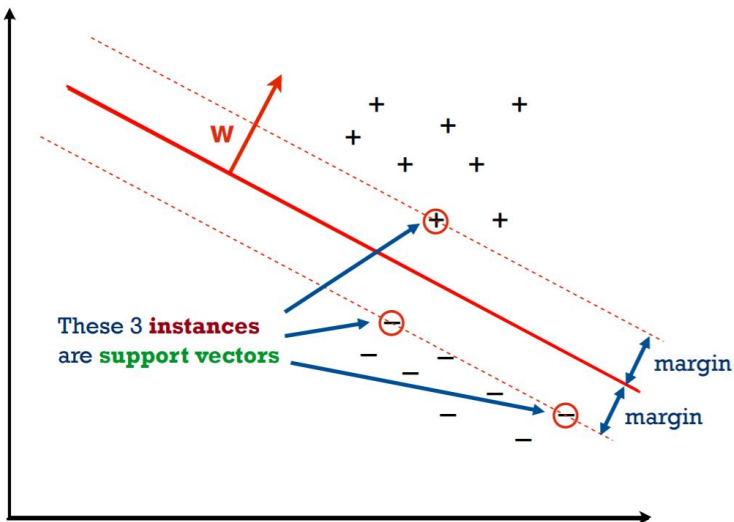
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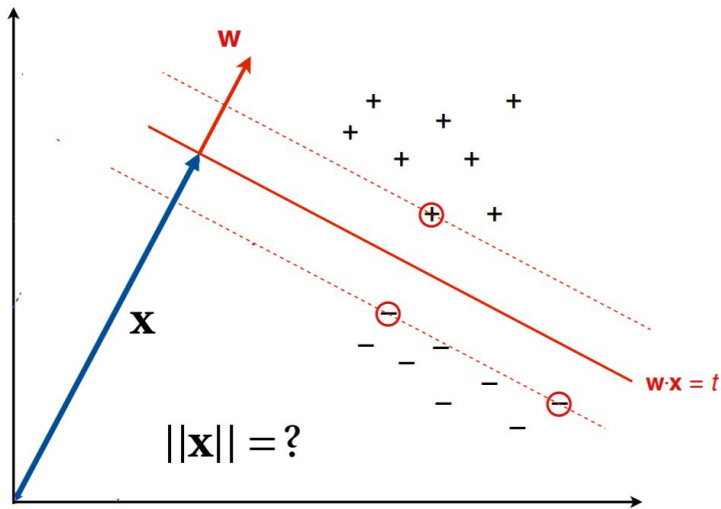
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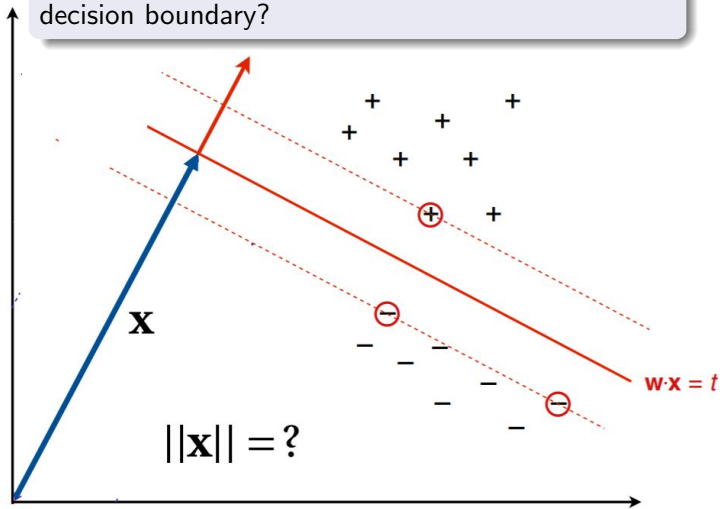


Calculating the margin



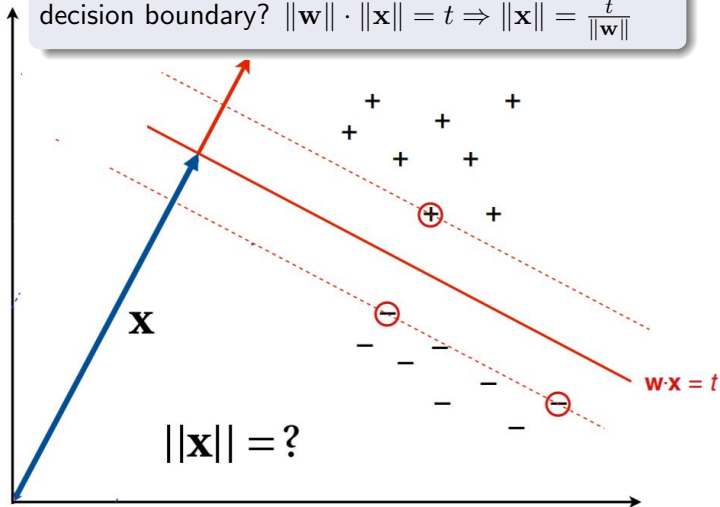
Calculating the margin

What is the distance between origin and the decision boundary?

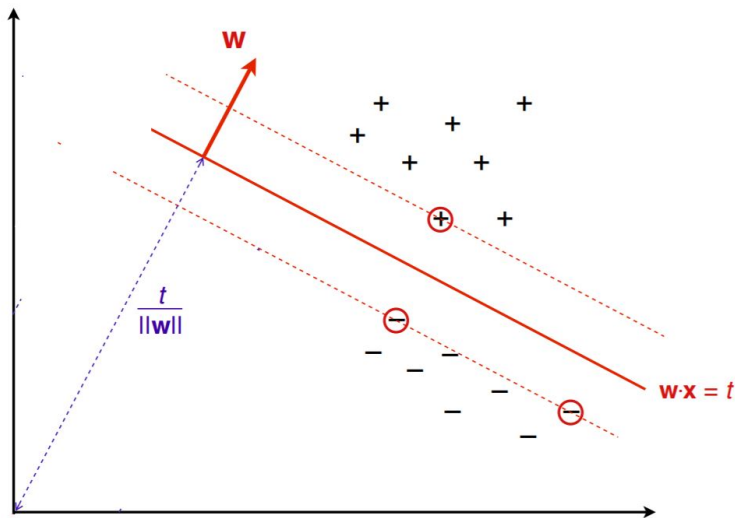


Calculating the margin

What is the distance between origin and the decision boundary? $\|\mathbf{w}\| \cdot \|\mathbf{x}\| = t \Rightarrow \|\mathbf{x}\| = \frac{t}{\|\mathbf{w}\|}$

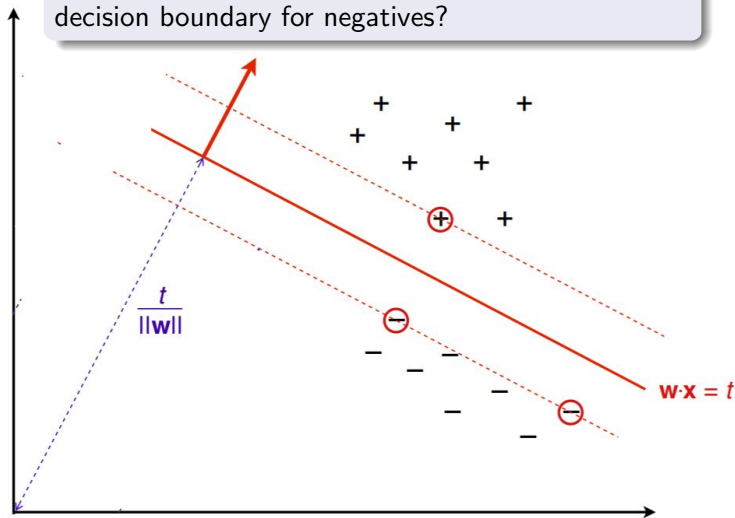


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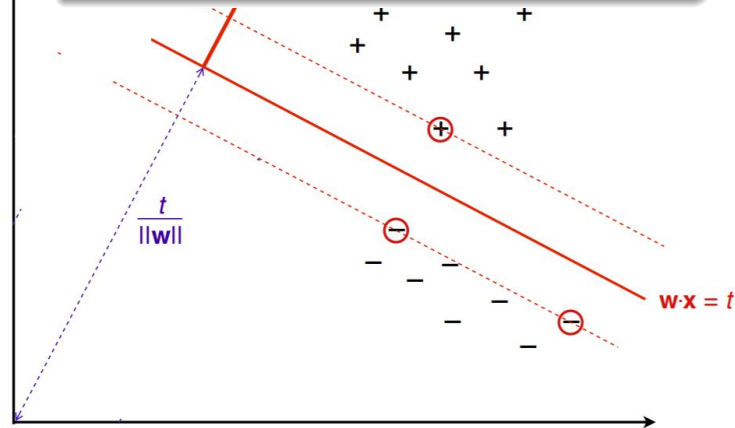
What is the distance between the origin and the decision boundary for negatives?



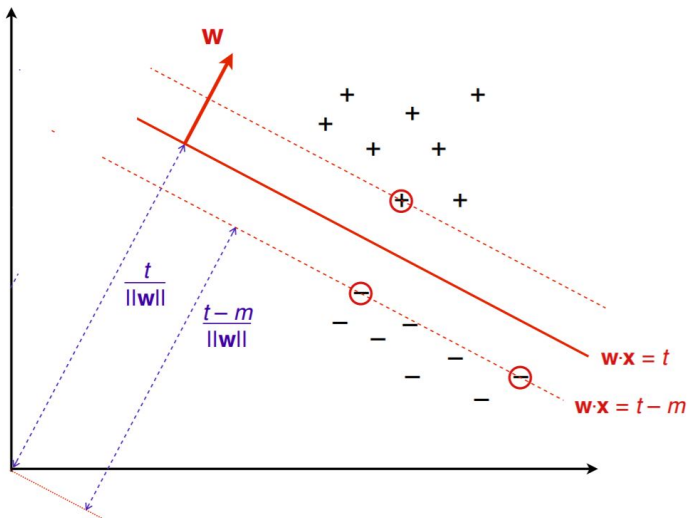
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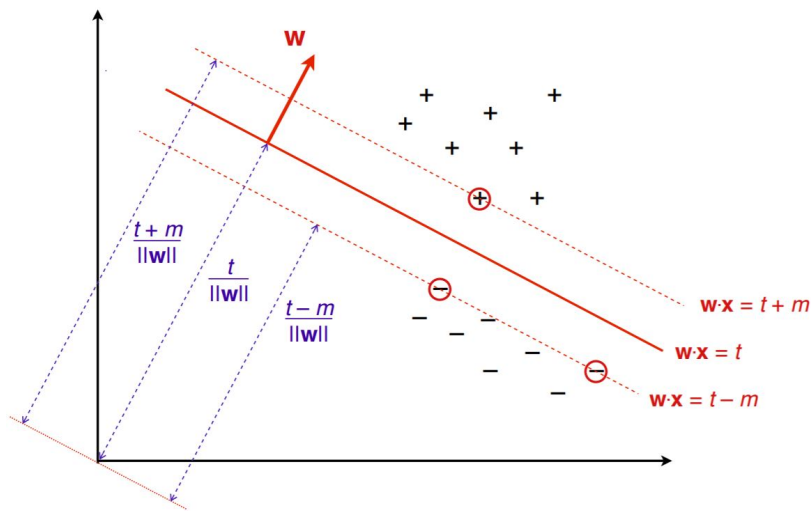
$$\|\mathbf{w}\| \cdot \|\mathbf{x}\| = t - m \Rightarrow \|\mathbf{x}\| = \frac{t-m}{\|\mathbf{w}\|}$$



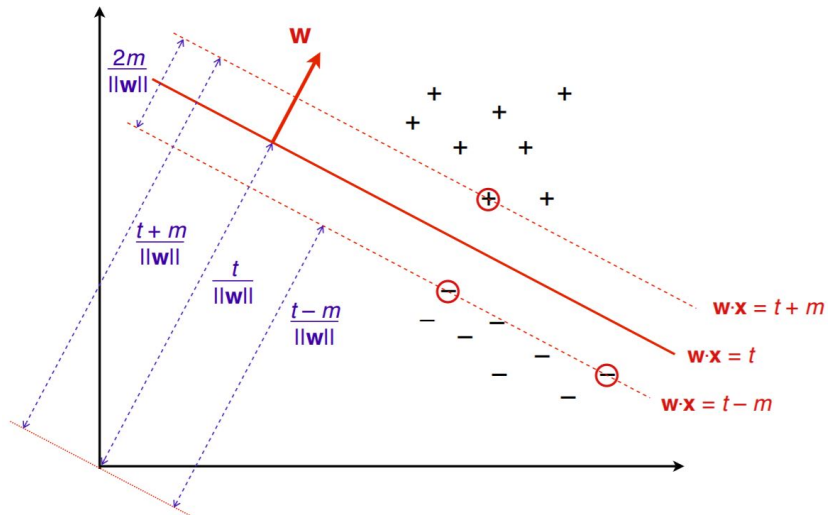
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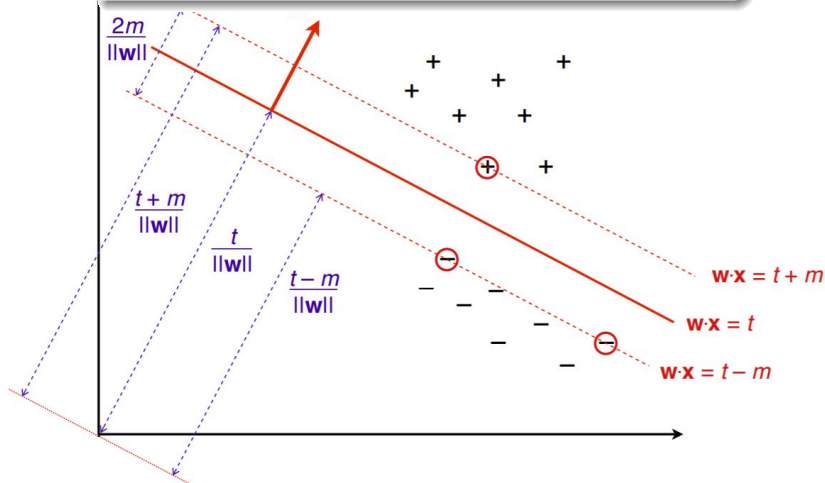


Key distances in SVM



Key distances in SVM

Since we are free to rescale t , w and m , it is customary to choose $m = 1$.



Optimization task in SVM

- Maximize the margin $\frac{1}{\|\mathbf{w}\|}$ such that
 - positives are at least by margin above the decision boundary:
 $\mathbf{w} \cdot \mathbf{x}_i \geq t + 1$
 - negatives are at least by margin below the decision boundary:
 $\mathbf{w} \cdot \mathbf{x}_i \leq t - 1$
- More conveniently and equivalently:

$$\mathbf{w}^*, t^* = \operatorname{argmin}_{\mathbf{w}, t} \frac{1}{2} \|\mathbf{w}\|^2$$

subject to $y_i(\mathbf{w} \cdot \mathbf{x}_i - t) \geq 1, \quad y_i \in \{-1, 1\}, \quad 1 \leq i \leq n$

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Suppose $y_i(\mathbf{w} \cdot \mathbf{x}_i - t) = 1$. What can we say about $\mathbf{x}_i$?

Hard-margin SVM

$$\mathbf{w}^*, t^* = \underset{\mathbf{w}, t}{\operatorname{argmin}} \frac{1}{2} \|\mathbf{w}\|^2$$

subject to $y_i(\mathbf{w} \cdot \mathbf{x}_i - t) \geq 1, \quad y_i \in \{-1, 1\}, \quad 1 \leq i \leq n$

To solve this optimization problem, first let's form the Lagrange function

$$\Lambda(\mathbf{w}, t, \alpha_1, \dots, \alpha_n) =$$

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Gradients of the Lagrangian

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Gradients of the Lagrangian

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$$\frac{\partial}{\partial \mathbf{w}} \Lambda(\mathbf{w}, t, \alpha_1, \dots, \alpha_n) = \mathbf{w} - \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i = 0$$

$$\mathbf{w} = \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i$$

Plugging back into the Lagrangian

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Applying KKT method on the Lagrangian

$$\Lambda(\alpha_1, \dots, \alpha_n) = -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j \mathbf{x}_i \cdot \mathbf{x}_j + \sum_{i=1}^n \alpha_i$$

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$$\text{subject to } \alpha_i \geq 0, \quad i = 1, \dots, n \text{ and } \sum_{i=1}^n \alpha_i y_i = 0$$

This task can be solved by quadratic optimisation solvers as we will see during the lab session.

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$$\mathbf{w}^*, t^* = \operatorname{argmin}_{\mathbf{w}, t} \frac{1}{2} \|\mathbf{w}\|^2$$

subject to $y_i(\mathbf{w} \cdot \mathbf{x}_i - t) \geq 1, \quad y_i \in \{-1, 1\}, \quad i = 1, \dots, n$

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- This was a **hard margin SVM** – we assumed that the classes are linearly separable
- **Soft margin SVM** can tolerate **margin errors**: cases where an instance is inside the margin or even at the wrong side of the decision boundary
- The idea is to introduce slack variables $\xi_i \geq 0$, one for each instance, measuring the amount of margin error (or equal to 0 if no error)

- The task is the following:

$$\mathbf{w}^*, t^*, \xi_i^* = \operatorname{argmin}_{\mathbf{w}, t, \xi_i} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i$$

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- Higher C usually results in more support vectors, hence C is a **complexity parameter**

Lagrangian function

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$$\frac{\partial}{\partial \xi_i} \Lambda(\mathbf{w}, t, \xi_i, \alpha_i, \beta_i) = C - \alpha_i - \beta_i = 0 \Rightarrow \alpha_i \leq C,$$

since $\beta_i \geq 0$

Summary of soft-margin SVM

$$\mathbf{w}^*, t^*, \xi_i^* = \operatorname{argmin}_{\mathbf{w}, t, \xi_i} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i$$

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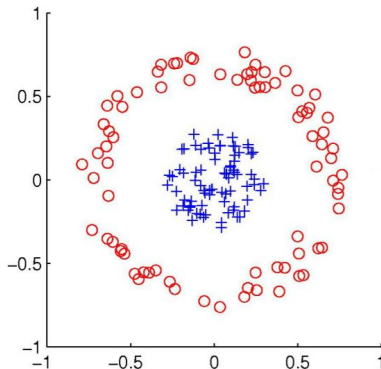
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- Con:** Works efficiently on relatively small datasets

Motivation for Kernels

- We know that KNN results in non-linear models (decision boundary is not a straight line / hyperplane)

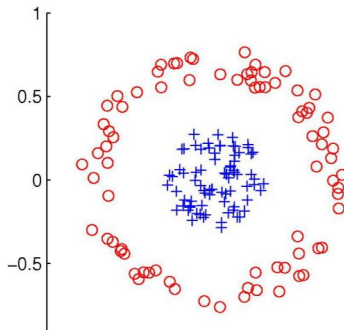
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It turns out that linear methods can also learn this separation! How?

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- Linear models are linear in the features

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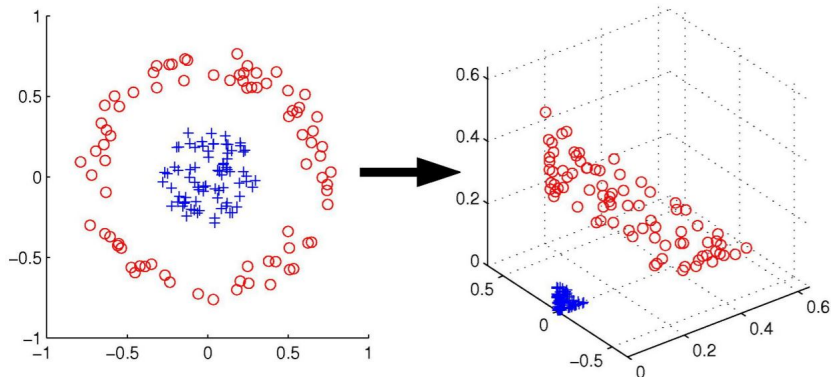
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- Linear models on the new features are nonlinear in the original features!
- This way we can fit non-linear models using linear model learning algorithms e.g.
 - if the hidden dependency is $y = x^3 + 3x^2 - x + 7$
 - Then if we introduce features x, x^2, x^3 then linear methods can learn this functional dependency

Non-linearity with linear models

Let's construct new features to make data linearly separable:

$$\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$(x_1, x_2) \rightarrow (z_1, z_2, z_3) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)$$



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- Let's see how $\phi(\mathbf{x})$ is used in SVM

SVM on constructed features

- Fitting (dual form soft-margin):

$\mathbf{x}_i \rightarrow \phi(\mathbf{x}_i)$ for $i = 1, \dots, n$

$$\alpha_1^*, \dots, \alpha_n^* = \operatorname{argmax}_{\alpha_1, \dots, \alpha_n} -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j \phi(\mathbf{x}_i) \cdot \phi(\mathbf{x}_j) + \sum_{i=1}^n \alpha_i$$

$$\text{subject to } 0 \leq \alpha_i \leq C, \quad i = 1, \dots, n \text{ and } \sum_{i=1}^n \alpha_i y_i = 0$$

$$\mathbf{w}^* = \sum_{i=1}^n \alpha_i^* y_i \phi(\mathbf{x}_i) \quad t^* = \mathbf{w}^* \cdot \phi(\mathbf{x}_j) - y_j = \sum_{i=1}^n \alpha_i^* y_i \phi(\mathbf{x}_i) \phi(\mathbf{x}_j) - y_j$$

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- Prediction:

$\mathbf{x} \rightarrow \phi(\mathbf{x})$

$$\hat{y} = \operatorname{sign}(\mathbf{w}^* \cdot \phi(\mathbf{x}) - t^*) = \operatorname{sign}\left(\sum_{i=1}^n \alpha_i^* y_i \phi(\mathbf{x}_i) \cdot \phi(\mathbf{x}) - t^*\right)$$

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- Dot product:

$$\phi(\mathbf{x}) \cdot \phi(\mathbf{x}') = \phi(x_1, x_2) \cdot \phi(x'_1, x'_2) =$$

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 - Use a modified method of calculating dot products $\mathbf{x} \cdot \mathbf{x}' \rightarrow \kappa(\mathbf{x}, \mathbf{x}')$
 - In our example the **kernel** is $\kappa(\mathbf{x}, \mathbf{x}') = (\mathbf{x} \cdot \mathbf{x}')^2$

- Fitting (soft-margin kernel SVM):

$$\alpha_1^*, \dots, \alpha_n^* = \operatorname{argmax}_{\alpha_1, \dots, \alpha_n} -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j \kappa(\mathbf{x}_i, \mathbf{x}_j) + \sum_{i=1}^n \alpha_i$$

$$\text{subject to } 0 \leq \alpha_i \leq C, \quad i = 1, \dots, n \text{ and } \sum_{i=1}^n \alpha_i y_i = 0$$

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- Prediction:

$$\hat{y} = \operatorname{sign} \left(\sum_{i=1}^n \alpha_i^* y_i \kappa(\mathbf{x}_i, \mathbf{x}) - t^* \right)$$

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- Mercer's theorem:
Such transformation ϕ exists if and only if $\kappa(\mathbf{x}, \mathbf{x}')$ is a positive semi-definite function

Positive semi-definiteness

Definition

Let $\kappa : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, then function $\kappa(\mathbf{x}, \mathbf{x}')$ is positive semi-definite, if for any $\mathbf{x}_1, \dots, \mathbf{x}_n$ the following matrix (**kernel matrix**) is positive semi-definite:

$$\begin{pmatrix} \kappa(\mathbf{x}_1, \mathbf{x}_1) & \kappa(\mathbf{x}_1, \mathbf{x}_2) & \dots & \kappa(\mathbf{x}_1, \mathbf{x}_n) \\ \kappa(\mathbf{x}_2, \mathbf{x}_1) & \kappa(\mathbf{x}_2, \mathbf{x}_2) & \dots & \kappa(\mathbf{x}_2, \mathbf{x}_n) \\ \vdots & \vdots & \ddots & \vdots \\ \kappa(\mathbf{x}_n, \mathbf{x}_1) & \kappa(\mathbf{x}_n, \mathbf{x}_2) & \dots & \kappa(\mathbf{x}_n, \mathbf{x}_n) \end{pmatrix}$$

Definition

A matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ is positive semi-definite, if for any vector $\mathbf{x} \in \mathbb{R}^n$

$$\mathbf{x}^T \mathbf{M} \mathbf{x} \geq 0$$

Checking these criteria might be non-trivial, often new kernels are constructed by combining existing ones

Well-known kernels

- Linear kernel (original space unchanged):

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- Gaussian kernel (also known as Radial Basis Function (RBF)) for any $\sigma > 0$ ($\gamma > 0$):

$$\kappa(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\sigma}\right) = \exp(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2)$$

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- Constructs all polynomials of features x_1, x_2 up to $d = 2$ degree

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Here the feature map is the following: $\phi(\mathbf{x}) = \left(e^{-\frac{\|\mathbf{x}\|^2}{2}} \frac{x_1^{n_1} \dots x_k^{n_k}}{\sqrt{n_1! \dots n_k!}} \right)_{\sum_{i=1}^k n_i = j}$

Construction of kernels

If κ_1, κ_2 are kernels, $a > 0$ and $f : \mathbb{X} \rightarrow \mathbb{R}$ then the following $\kappa(\mathbf{x}, \mathbf{x}')$ are kernels as well:

$$\kappa(\mathbf{x}, \mathbf{x}') = a\kappa_1(\mathbf{x}, \mathbf{x}')$$

$$\kappa(\mathbf{x}, \mathbf{x}') = \kappa_1(\mathbf{x}, \mathbf{x}')\kappa_2(\mathbf{x}, \mathbf{x}')$$

$$\kappa(\mathbf{x}, \mathbf{x}') = \kappa_1(\mathbf{x}, \mathbf{x}') + \kappa_2(\mathbf{x}, \mathbf{x}')$$

$$\kappa(\mathbf{x}, \mathbf{x}') = f(\mathbf{x})f(\mathbf{x}')$$

What have we learned today?

- ✓ Convex Optimization
- ✓ Support Vector Machine (SVM)
- ✓ Hard-margin SVM
- ✓ Soft-margin SVM
- ✓ Kernel Methods